Explainable Machine Learning and Grounded Policy Retrieval Framework for Food Inflation Forecasting and Economic Early Warning in Bangladesh

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Keywords— *Food Inflation Nowcasting, Explainable Machine Learning, Economic Early Warning, Conformal Prediction, Grounded Policy Retrieval.*

1. Research Area

Applied Machine Learning, Computational Macroeconomics, Empirical Time-Series Forecasting, and Retrieval-Augmented Natural Language Processing.

2. Goals/Objectives

This research addresses severe macroeconomic instability and staple food price shocks in developing, import-dependent economies, focusing specifically on Bangladesh. The primary operational challenge stems from institutional publication friction: official Consumer Price Index (CPI) bulletins and foreign exchange reserve releases from the Bangladesh Bureau of Statistics (BBS) and Bangladesh Bank are delayed by four to six weeks. Consequently, government market cooling interventions and import subsidy programs remain reactive rather than anticipatory. To resolve this acute informational deficit, this research establishes four concrete objectives: (i) constructing an explainable multi-indicator machine learning framework to nowcast short-horizon food inflation trajectory; (ii) deploying an adaptive rolling walk-forward conformal prediction engine that delivers finite-sample valid retail commodity price bands; (iii) developing an unsupervised anomaly detection system that generates actionable early warnings months prior to systemic macroeconomic stress; and (iv) engineering an evidence-grounded policy advisory retrieval framework that synthesizes central bank policy directives without generative hallucination.

3. Methodology

The empirical architecture is anchored on a genuine macroeconomic panel spanning January 2010 to October 2026 (6,144 daily observations, 42 macroeconomic and global commodity indicators) coupled with an authentic monthly macroeconomic panel (202 continuous months, zero synthetic interpolation). Institutional release schedules (BBS CPI on Day 10 of M+1, Bangladesh Bank reserves on Day 7 of M+1, WFP retail surveys on Day 15 of M) are strictly enforced via step-function forward fills to guarantee zero look-ahead data leakage. Global high-frequency shocks (Brent crude oil, CBOT wheat futures, US Dollar Index, NYMEX natural gas, and gold) are ingested continuously. For food inflation nowcasting, an Extreme Gradient Boosting (XGBoost) model is trained with temporal expanding windows and benchmarked against linear Ridge regression and naive random walk baselines using Harvey-Leybourne-Newbold corrected Diebold-Mariano tests. Feature attributions are extracted via TreeSHAP to quantify international import price pass-through dynamics. For retail staple commodities (coarse rice, lentils, wheat flour), a rolling walk-forward adaptive conformal prediction engine is formulated over a 730-day calibration window, generating distribution-free 80% coverage intervals on unseen test data. For crisis surveillance, an Isolation Forest is trained strictly on pre-2022 historical controls and paired with a reset-capable Cumulative Sum (CUSUM) filter to detect persistent structural shocks. Finally, a ChromaDB-backed Retrieval-Augmented Generation (RAG) system ingests authentic monetary policy statements to answer domain policy queries.

4. Results and Analysis

Empirical evaluations confirm that non-linear machine learning substantially outperforms traditional linear econometric baselines in capturing international shock transmissions into domestic prices. In monthly forecasting (Table I), XGBoost reduces root mean square error (RMSE) relative to the linear Ridge baseline by 19.99% at the t+1 horizon (1.171% vs. 1.464%, Diebold-Mariano statistic = 1.82, $p = 0.068$) and by 32.50% at the t+3 horizon (1.740% vs. 2.578%, DM = 1.90, $p = 0.058$). In daily 30-day forward nowcasting, XGBoost achieves decisive statistical significance against Ridge (RMSE 1.158% vs. 2.457%, a 52.87% error reduction; DM = 3.74, $p = 0.00019$), while demonstrating parity with naive persistence, confirming strong domestic price stickiness. TreeSHAP feature attributions prove that global crude oil, CBOT wheat, and the US Dollar Index explain over 50% of aggregate forecast variance, validating trade pass-through theory.

Forecast Horizon	Naive Baseline	Linear Ridge	Random Forest	Proposed XGBoost	Gain vs. Ridge	DM Stat (p-value)
t+1 (1 Month Ahead)	1.160%	1.464%	1.087%	**1.171%**	**19.99%** ↓	1.82 (p = 0.068)
t+3 (3 Months Ahead)	1.757%	2.578%	1.725%	**1.740%**	**32.50%** ↓	1.90 (p = 0.058)
30-Day Forward Daily	1.137%	2.457%	1.215%	**1.158%**	**52.87%** ↓	**3.74 (p < 0.001)**

Table I: Out-of-Sample Forecasting Performance & Diebold-Mariano Tests Across Horizons

In retail commodity interval prediction, the rolling conformal engine yields 85.1% empirical out-of-sample coverage for coarse rice, 74.8% for lentils, and 75.1% for flour across 1,366 unseen test days. Crucially, the unsupervised anomaly detection architecture successfully detected both out-of-sample economic crisis episodes—the 2022 Russia-Ukraine foreign exchange drain (80 days in advance) and the 2024 political and inflation shock (90 days in advance)—delivering an average operational early warning lead time of 85 days (2.8 months) with 100% episode recall. The grounded policy RAG achieved an 82.0% mean grounding score across held-out central bank queries. The complete framework is open-source and deployed as a real-time policy dashboard.

5. References

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